

Gihyeok (Peter) Lee

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PROFESSIONAL SUMMARY

Nanotechnology Engineering student with experience in polymer formulation, materials characterization, and additive manufacturing. Led optimization and scale-up of high-performance SLS powders and conducted mechanical validation to support material qualification. Developed automated image analysis pipelines to improve experimental efficiency and data reliability.

EDUCATION

University of Waterloo

Honours Nanotechnology Engineering, Co-operative Program, BASc.

Waterloo, ON

Sept. 2023 – April 2028

EXPERIENCE

SLS Materials Intern

September 2025 – April 2026

Formlabs

Boston, MA

- Led material optimization of high-performance polyamide powder, increasing elongation at break from <2% to 8% through base polymer and additive reformulation, enabling repositioning from specialty to higher volume general-purpose use
- Led 16× scale-up of powder blending processes from benchtop to industrial mixers for next-generation SLS powders, ensuring formulation consistency and enabling production-scale manufacturing
- Contributed to the launch of a high-ductility nylon powder for next-generation SLS platforms by identifying conditions for improved part quality and post processing by increasing defect free part yield by 25%

Undergraduate Research Assistant

February 2025 – April 2025

Maikwa Lab/University of Toronto

Toronto, ON

- Designed and optimized an Automated Mucoadhesion Assay using 3D-printed components, achieving 70% reduction in particle size and 90% reduction in processing time
- Developed and tested 7 iterative grinder head designs using SolidWorks, optimizing for minimal average particle size and low polydispersity; validated results using ImageJ analysis of microscope data.
- Performed quantitative NMR experiments to optimize UV curing durations, precisely correlating crosslinking efficiency with photoinitiator exposure time.

EXTRACURRICULARS

HackerFab Process Core Member

May 2026 – Present

- Standardized photoresist spin coating on diced silicon by defining dispense, spin, bake, and handling parameters in a team SOP. Reduced patterning failures by 70%
- Fabricating MOS capacitors using thermal oxidation, photolithography, wet etching, and aluminum sputtering
- Building a low-cost laser-scanning microscope using a DVD optical pickup and piezoelectric stage; evaluating photodiode readout and stage-control methods

Automated Particle Detection and Measurement System | *OpenCV, numpy, pandas*

April 2025

- Developed a Python-based tool using OpenCV to process microscope images and automatically detect particles
- Implemented Otsu's thresholding method for binary image conversion, followed by contour detection to identify particle boundaries
- Exported particle measurement data (diameters) into an Excel spreadsheet for further analysis

TECHNICAL SKILLS

Languages: Java, Python, Matlab

Software: SolidWorks, ImageJ, Trios, Onshape

Lab Equipment: SEM, DSC, TGA, qNMR, Powder Rheology, Optical Microscope, Tube Furnace, Sputter Coater, Spin Coater